

Reverse-check review package — 368sextets_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	368sextets/repair/final.fr
original model name	368sextets_gen (hidden from the agent)
paper	368sextets/text/2110.11359.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LSextetKin (:=)

```
sFubar[ss,kk].(I Ga[mu,ss,rr].DC[sFu[rr,kk],mu] - MFu sFu[ss,kk]) + sFdbar[ss,kk].(I
  ↪ Ga[mu,ss,rr].DC[sFd[rr,kk],mu] - MFd sFd[ss,kk]) + DC[sSubar[kk], mu] DC[sSu[kk],mu] -
  ↪ MSu^2 sSubar[kk] sSu[kk] + DC[sSdbar[kk], mu] DC[sSd[kk],mu] - MSd^2 sSdbar[kk] sSd[kk]
```

LFu (:=)

```
CFu[mm] I Sqrt[2] K6[kk,yy,zz] T[aa,zz,xx] K3bar[ii,xx,yy] sFubar[ss,kk].(I/2)(Ga[mu,ss,rr]
  ↪ .Ga[nu,rr,tt]-Ga[nu,ss,rr].Ga[mu,rr,tt]).CC[uR][tt,mm,ii] FS[G,mu,nu,aa] + HC[CFu[mm]]
  ↪ (-I Sqrt[2]) K3[ii,xx,yy] T[aa,xx,zz] K6bar[kk,zz,yy] HC[sFubar[ss,kk].(I/2)(Ga[mu,ss,rr]
  ↪ ].Ga[nu,rr,tt]-Ga[nu,ss,rr].Ga[mu,rr,tt]).CC[uR][tt,mm,ii]] FS[G,mu,nu,aa] + CFBu[mm] I
  ↪ Sqrt[2] K6[kk,yy,zz] T[aa,zz,xx] K3bar[ii,xx,yy] sFubar[ss,kk].CC[uR][ss,mm,ii]
  ↪ FS[G,mu,nu,aa] FS[B,mu,nu] + HC[CFBu[mm]] (-I Sqrt[2]) K3[ii,xx,yy] T[aa,xx,zz]
  ↪ K6bar[kk,zz,yy] HC[sFubar[ss,kk].CC[uR][ss,mm,ii]] FS[G,mu,nu,aa] FS[B,mu,nu]
```

LFd (:=)

```
CFd[mm] I Sqrt[2] K6[kk,yy,zz] T[aa,zz,xx] K3bar[ii,xx,yy] sFdbar[ss,kk].(I/2)(Ga[mu,ss,rr]
  ↪ .Ga[nu,rr,tt]-Ga[nu,ss,rr].Ga[mu,rr,tt]).CC[dR][tt,mm,ii] FS[G,mu,nu,aa] + HC[CFd[mm]]
  ↪ (-I Sqrt[2]) K3[ii,xx,yy] T[aa,xx,zz] K6bar[kk,zz,yy] HC[sFdbar[ss,kk].(I/2)(Ga[mu,ss,rr]
  ↪ ].Ga[nu,rr,tt]-Ga[nu,ss,rr].Ga[mu,rr,tt]).CC[dR][tt,mm,ii]] FS[G,mu,nu,aa] + CFBd[mm] I
  ↪ Sqrt[2] K6[kk,yy,zz] T[aa,zz,xx] K3bar[ii,xx,yy] sFdbar[ss,kk].CC[dR][ss,mm,ii]
  ↪ FS[G,mu,nu,aa] FS[B,mu,nu] + HC[CFBd[mm]] (-I Sqrt[2]) K3[ii,xx,yy] T[aa,xx,zz]
  ↪ K6bar[kk,zz,yy] HC[sFdbar[ss,kk].CC[dR][ss,mm,ii]] FS[G,mu,nu,aa] FS[B,mu,nu]
```

LSu (:=)

```
CSu[mm,ll] I Sqrt[2] K6[kk,yy,zz] T[aa,zz,xx] K3bar[ii,xx,yy] sSubar[kk]
  ↪ lRbar[ss,ll].(I/2)(Ga[mu,ss,rr].Ga[nu,rr,tt]-Ga[nu,ss,rr].Ga[mu,rr,tt]).CC[uR][tt,mm,ii]
  ↪ FS[G,mu,nu,aa] + HC[CSu[mm,ll]] (-I Sqrt[2]) K3[ii,xx,yy] T[aa,xx,zz] K6bar[kk,zz,yy]
  ↪ HC[sSubar[kk]
  ↪ lRbar[ss,ll].(I/2)(Ga[mu,ss,rr].Ga[nu,rr,tt]-Ga[nu,ss,rr].Ga[mu,rr,tt]).CC[uR][tt,mm,ii]
  ↪ FS[G,mu,nu,aa]]
```

LSd (:=)

```
CSd[mm,ll] I Sqrt[2] K6[kk,yy,zz] T[aa,zz,xx] K3bar[ii,xx,yy] sSdbar[kk]
  ↳ lRbar[ss,ll] . (I/2) (Ga[mu,ss,rr] . Ga[nu,rr,tt] - Ga[nu,ss,rr] . Ga[mu,rr,tt]) . CC[dR][tt,mm,ii]
  ↳ FS[G,mu,nu,aa] + HC[CSd[mm,ll]] (-I Sqrt[2]) K3[ii,xx,yy] T[aa,xx,zz] K6bar[kk,zz,yy]
  ↳ HC[sSdbar[kk]]
  ↳ lRbar[ss,ll] . (I/2) (Ga[mu,ss,rr] . Ga[nu,rr,tt] - Ga[nu,ss,rr] . Ga[mu,rr,tt]) . CC[dR][tt,mm,ii]
  ↳ FS[G,mu,nu,aa]
```

LTot (:=)

LSextetKin + LFu + LFd + LSu + LSd

Blank-slate reconstruction

Reconstructed Model From sanitized.fr

Notation

Generation indices are m, ℓ . Color triplet indices are i, x, y, z , adjoint color index is a , and sextet index is $k = 1, \dots, 6$. Repeated indices are summed.

The file defines the color tensor

$$\mathcal{C}_{ki}^a \equiv \sqrt{2} K_{6kyz} (T^a)_{zx} \bar{K}_{3ixy},$$

with hermitian-conjugate tensor

$$\mathcal{C}_{ki}^{a*} = \sqrt{2} K_{3ixy} (T^a)_{xz} \bar{K}_{6kzy}.$$

The FeynRules expression

$$\frac{i}{2} (\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu)$$

is written as

$$\sigma^{\mu\nu}.$$

Charge conjugation is

$$q_R^c \equiv C \bar{q}_R^T.$$

The fields are all $SU(2)_L$ singlets. For a field Φ_k in the sextet representation with hypercharge Y_Φ ,

$$D_\mu \Phi_k = \partial_\mu \Phi_k + ig_s G_\mu^a (T_6^a)_{kk'} \Phi_{k'} + ig' Y_\Phi B_\mu \Phi_k.$$

For the conjugate field,

$$D_\mu \Phi_k^\dagger = \partial_\mu \Phi_k^\dagger - ig_s G_\mu^a \Phi_{k'}^\dagger (T_6^a)_{k'k} - ig' Y_\Phi B_\mu \Phi_k^\dagger.$$

The field strengths are

$$G_{\mu\nu}^a = FS[G, \mu, \nu, a], \quad B_{\mu\nu} = FS[B, \mu, \nu].$$

Lagrangian

LSextetKin

$$\mathcal{L}_{\text{LSextetKin}} \supset \bar{s} F u_k (i \gamma^\mu D_\mu - M_{Fu}) s F u_k.$$

$$\mathcal{L}_{\text{LSextetKin}} \supset \bar{s} F d_k (i \gamma^\mu D_\mu - M_{Fd}) s F d_k.$$

$$\mathcal{L}_{\text{LSextetKin}} \supset (D_\mu s S u_k^\dagger) (D^\mu s S u_k) - M_{Su}^2 s S u_k^\dagger s S u_k.$$

$$\mathcal{L}_{\text{LSextetKin}} \supset (D_\mu s S d_k^\dagger)(D^\mu s S d_k) - M_{Sd}^2 s S d_k^\dagger s S d_k.$$

The covariant derivatives use the following hypercharges:

$$Y_{sFu} = -\frac{2}{3}, \quad Y_{sFd} = +\frac{1}{3}, \quad Y_{sSu} = +\frac{1}{3}, \quad Y_{sSd} = +\frac{4}{3}.$$

LFu

$$\mathcal{L}_{\text{LFu}} \supset i C_{Fu}^m \mathcal{C}_{ki}^a \bar{s} F u_k \sigma^{\mu\nu} (u_R^c)_{mi} G_{\mu\nu}^a + \text{h.c.}$$

$$\mathcal{L}_{\text{LFu}} \supset i C_{FBu}^m \mathcal{C}_{ki}^a \bar{s} F u_k (u_R^c)_{mi} G_{\mu\nu}^a B^{\mu\nu} + \text{h.c.}$$

where

$$C_{Fu}^m = C F u R_m + i C F u I_m, \quad C_{FBu}^m = C F B u R_m + i C F B u I_m.$$

LFd

$$\mathcal{L}_{\text{LFd}} \supset i C_{Fd}^m \mathcal{C}_{ki}^a \bar{s} F d_k \sigma^{\mu\nu} (d_R^c)_{mi} G_{\mu\nu}^a + \text{h.c.}$$

$$\mathcal{L}_{\text{LFd}} \supset i C_{FBd}^m \mathcal{C}_{ki}^a \bar{s} F d_k (d_R^c)_{mi} G_{\mu\nu}^a B^{\mu\nu} + \text{h.c.}$$

where

$$C_{Fd}^m = C F d R_m + i C F d I_m, \quad C_{FBd}^m = C F B d R_m + i C F B d I_m.$$

LSu

$$\mathcal{L}_{\text{LSu}} \supset i C_{Su}^{m\ell} \mathcal{C}_{ki}^a s S u_k^\dagger \bar{\ell}_{R\ell} \sigma^{\mu\nu} (u_R^c)_{mi} G_{\mu\nu}^a + \text{h.c.}$$

where

$$C_{Su}^{m\ell} = C S u R_{m\ell} + i C S u I_{m\ell}.$$

LSd

$$\mathcal{L}_{\text{LSd}} \supset i C_{Sd}^{m\ell} \mathcal{C}_{ki}^a s S d_k^\dagger \bar{\ell}_{R\ell} \sigma^{\mu\nu} (d_R^c)_{mi} G_{\mu\nu}^a + \text{h.c.}$$

where

$$C_{Sd}^{m\ell} = C S d R_{m\ell} + i C S d I_{m\ell}.$$

LTot

$$\mathcal{L}_{\text{LTot}} = \mathcal{L}_{\text{LSextetKin}} + \mathcal{L}_{\text{LFu}} + \mathcal{L}_{\text{LFd}} + \mathcal{L}_{\text{LSu}} + \mathcal{L}_{\text{LSd}}.$$

Field Table

Symbol	Spin	$SU(3)_c$ rep	$SU(2)_L$ rep	$U(1)_Y$	Electric charge	Self-conjugate	Mass
sFu	1/2 Dirac fermion	6	1	-2/3	-2/3	No	$M_{Fu} = 500$
sFd	1/2 Dirac fermion	6	1	+1/3	+1/3	No	$M_{Fd} = 500$
sSu	scalar	6	1	+1/3	+1/3	No	$M_{Su} = 500$

Symbol	Spin	$SU(3)_c$ rep	$SU(2)_L$ rep	$U(1)_Y$	Electric charge	Self-conjugate	Mass
sSd	scalar	6	1	+4/3	+4/3	No	$M_{Sd} = 500$

The antiparticles transform in the conjugate color representation $\bar{\mathbf{6}}$ with opposite gauge charges.

External Parameters

External parameter	Default	Multiplies	Physical meaning
CFBuR [m]	0.1	Real part of C_{FBu}^m in LFu	Coupling of $\bar{s}Fu u_R^c G_{\mu\nu} B^{\mu\nu}$
CFBuI [m]	0	Imaginary part of C_{FBu}^m in LFu	CP-odd/complex phase component of the same coupling
CFBdR [m]	0.1	Real part of C_{FBd}^m in LFd	Coupling of $\bar{s}Fd d_R^c G_{\mu\nu} B^{\mu\nu}$
CFBdI [m]	0	Imaginary part of C_{FBd}^m in LFd	CP-odd/complex phase component of the same coupling
CFuR [m]	0.1	Real part of C_{Fu}^m in LFu	Chromomagnetic-type coupling $\bar{s}Fu \sigma^{\mu\nu} u_R^c G_{\mu\nu}^a$
CFuI [m]	0	Imaginary part of C_{Fu}^m in LFu	CP-odd/complex phase component of the same coupling
CFdR [m]	0.1	Real part of C_{Fd}^m in LFd	Chromomagnetic-type coupling $\bar{s}Fd \sigma^{\mu\nu} d_R^c G_{\mu\nu}^a$
CFdI [m]	0	Imaginary part of C_{Fd}^m in LFd	CP-odd/complex phase component of the same coupling
CSuR [m, 1]	0.1	Real part of $C_{Su}^{m\ell}$ in LSu	Coupling of $sSu^\dagger \bar{\ell}_R \sigma^{\mu\nu} u_R^c G_{\mu\nu}^a$
CSuI [m, 1]	0	Imaginary part of $C_{Su}^{m\ell}$ in LSu	CP-odd/complex phase component of the same coupling
CSdR [m, 1]	0.1	Real part of $C_{Sd}^{m\ell}$ in LSd	Coupling of $sSd^\dagger \bar{\ell}_R \sigma^{\mu\nu} d_R^c G_{\mu\nu}^a$
CSdI [m, 1]	0	Imaginary part of $C_{Sd}^{m\ell}$ in LSd	CP-odd/complex phase component of the same coupling

Physics Summary

The file encodes four new color-sextet, electroweak-singlet states: two Dirac fermions with charges $-2/3$ and $+1/3$, and two complex scalars with charges $+1/3$ and $+4/3$. Their interactions are higher-dimensional operators coupling them to right-handed charge-conjugated up- or down-type quarks, gluon field strengths, and in two fermion operators also the hypercharge field strength.

The model mediates processes involving color-sextet resonances connected to quarks plus gluons, and scalar sextet interactions involving a charged right-handed lepton, a charge-conjugated right-handed quark, and a gluon. These operators can generate exotic colored-particle production and decays into quark-gluon or lepton-quark-gluon final states, with generation-dependent complex couplings.

Paper cross-check

Paper Model References

The model targeted by the reconstruction is defined in **Section III, “Sextets at the LHC”**. The explicit fermion Lagrangian is **Eq. (12)**, the scalar Lagrangian is **Eq. (13)**, and the exotic field quantum numbers are summarized in **Table X**. The relevant operator catalog entry is **Table IX**, based on the color invariant $3 \otimes 6 \otimes 8$. The color Clebsch-Gordan conventions for J_{sia} and $\bar{J}_{sai}$ are given in **Appendix A**, especially **Eqs. (A15)–(A18)**.

paper term (eq. ref)	reconstruction term	verdict	notes
$\Psi_u \sim (6, 1, -2/3), L = 0$ (Table X)	sFu : Dirac fermion, (6, 1, $-2/3$), charge $-2/3$	agree	Gauge quantum numbers and spin agree. Reconstruction omits the paper’s explicit lepton-number assignment $L = 0$.
$\Psi_d \sim (6, 1, +1/3), L = 0$ (Table X)	sFd : Dirac fermion, (6, 1, $+1/3$), charge $+1/3$	agree	Gauge quantum numbers and spin agree. Reconstruction omits $L = 0$.
$\Phi_u \sim (6, 1, +1/3), L = -1$ (Table X)	sSu : scalar, (6, 1, $+1/3$), charge $+1/3$	agree	Gauge quantum numbers and spin agree. Reconstruction omits the scalar lepton number $L = -1$.
$\Phi_d \sim (6, 1, +4/3), L = -1$ (Table X)	sSd : scalar, (6, 1, $+4/3$), charge $+4/3$	agree	Gauge quantum numbers and spin agree. Reconstruction omits $L = -1$.
$\bar{\Psi}_q (i\bar{D} - m_{\Psi_q}) \Psi_q$ for $q \in \{u, d\}$ (Eq. 12)	$s\bar{F}u(i\gamma^\mu D_\mu - M_{Fu})sFu,$ $s\bar{F}d(i\gamma^\mu D_\mu - M_{Fd})sFd$	agree	Same Dirac kinetic and mass structure. Reconstruction names separate masses and gives implementation defaults later.
$(D_\mu \Phi_q)^\dagger D^\mu \Phi_q - m_{\Phi_q}^2 \Phi_q^\dagger \Phi_q$ for $q \in \{u, d\}$ (Eq. 13)	$(D_\mu sSu^\dagger)(D^\mu sSu) - M_{Su}^2 sSu^\dagger sSu$, similarly for sSd	agree	Same complex scalar kinetic and mass structure.
$J_{sia}, \bar{J}_{sai}$ color contraction for $3 \otimes 6 \otimes 8$ (Table IX , App. A , Eqs. A15–A18)	$\mathcal{C}_{ki}^a = \sqrt{2} K_{6kyz}(T^a)_{zx} \bar{K}_{3ixy}$, plus conjugate	agree	This appears to be the implementation form of the paper’s J tensor using the relation to K , L , and t_3^a in Eq. (A18). Overall phase/index placement should be checked against the implementation convention.

paper term (eq. ref)	reconstruction term	verdict	notes
$\frac{1}{\Lambda_{\Psi_u}^3} [\kappa_u^I J_{sia} (u_{RI}^c \sigma_{\mu\nu} \Psi_{us}) G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} u_k \sigma^{\mu\nu} (u_R^c)_{mi} G_{\mu\nu}^a]$ (Eq. 12)	$\frac{1}{\Lambda_{\Psi_u}^3} [\kappa_u^I J_{sia} (u_{RI}^c \sigma_{\mu\nu} \Psi_{us}) G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} u_k \sigma^{\mu\nu} (u_R^c)_{mi} G_{\mu\nu}^a]$ h.c.	disagree	Field content, chirality, gluon field strength, and sextet/up-quark assignment match up to Hermitian conjugation and Dirac-bilinear ordering. Reconstruction does not show the paper's $1/\Lambda_{\Psi_u}$ coefficient structure.
$\frac{1}{\Lambda_{\Psi_d}^3} [\kappa_d^I J_{sia} (d_{RI}^c \sigma_{\mu\nu} \Psi_{ds}) G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} d_k \sigma^{\mu\nu} (d_R^c)_{mi} G_{\mu\nu}^a]$ (Eq. 12)	$\frac{1}{\Lambda_{\Psi_d}^3} [\kappa_d^I J_{sia} (d_{RI}^c \sigma_{\mu\nu} \Psi_{ds}) G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} d_k \sigma^{\mu\nu} (d_R^c)_{mi} G_{\mu\nu}^a]$ h.c.	disagree	Same physics content as the paper operator, but the explicit EFT suppression $1/\Lambda_{\Psi_d}$ is absent or absorbed without being stated.
$\frac{1}{\Lambda_{\Psi_u B}^3} [\kappa_u^I J_{sia} (u_{RI}^c \Psi_{us}) B_{\mu\nu} G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} u_k (u_R^c)_{mi} G_{\mu\nu}^a B^{\mu\nu}]$ (Eq. 12)	$\frac{1}{\Lambda_{\Psi_u B}^3} [\kappa_u^I J_{sia} (u_{RI}^c \Psi_{us}) B_{\mu\nu} G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} u_k (u_R^c)_{mi} G_{\mu\nu}^a B^{\mu\nu}]$ h.c.	disagree	Correct fields and no $\sigma_{\mu\nu}$, matching the scalar Lorentz bilinear structure in Eq. (12), but reconstruction omits or absorbs the required $1/\Lambda^3$.
$\frac{1}{\Lambda_{\Psi_d B}^3} [\kappa_d^I J_{sia} (d_{RI}^c \Psi_{ds}) B_{\mu\nu} G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} d_k (d_R^c)_{mi} G_{\mu\nu}^a B^{\mu\nu}]$ (Eq. 12)	$\frac{1}{\Lambda_{\Psi_d B}^3} [\kappa_d^I J_{sia} (d_{RI}^c \Psi_{ds}) B_{\mu\nu} G^{\mu\nu\alpha} C_{Fda}^{m\alpha} \bar{F} d_k (d_R^c)_{mi} G_{\mu\nu}^a B^{\mu\nu}]$ h.c.	disagree	Same operator content, but the EFT coefficient/cutoff structure differs from the paper unless C_{FBd} implicitly includes $1/\Lambda^3$.
$\frac{1}{\Lambda_{\Phi_u}^2} [\lambda_u^{XI} J_{sia} \Phi_{us} (u_{RI}^c \sigma_{\mu\nu} \ell_{RX}^c C_{Sd}^{m\alpha} C_{kt}^{\alpha} \bar{F} u_k \sigma^{\mu\nu} (u_R^c)_{mi} G_{\mu\nu}^a)]$ (Eq. 13)	$\frac{1}{\Lambda_{\Phi_u}^2} [\lambda_u^{XI} J_{sia} \Phi_{us} (u_{RI}^c \sigma_{\mu\nu} \ell_{RX}^c C_{Sd}^{m\alpha} C_{kt}^{\alpha} \bar{F} u_k \sigma^{\mu\nu} (u_R^c)_{mi} G_{\mu\nu}^a)]$ h.c.	disagree	Reconstruction uses the Hermitian-conjugate-looking form with $sSu^\dagger$, which is acceptable because the paper includes H.c.; however, the explicit $1/\Lambda_{\Phi_u}^2$ suppression is missing or absorbed.
$\frac{1}{\Lambda_{\Phi_d}^2} [\lambda_d^{XI} J_{sia} \Phi_{ds} (d_{RI}^c \sigma_{\mu\nu} \ell_{RX}^c C_{Sd}^{m\alpha} C_{kt}^{\alpha} \bar{F} d_k \sigma^{\mu\nu} (d_R^c)_{mi} G_{\mu\nu}^a)]$ (Eq. 13)	$\frac{1}{\Lambda_{\Phi_d}^2} [\lambda_d^{XI} J_{sia} \Phi_{ds} (d_{RI}^c \sigma_{\mu\nu} \ell_{RX}^c C_{Sd}^{m\alpha} C_{kt}^{\alpha} \bar{F} d_k \sigma^{\mu\nu} (d_R^c)_{mi} G_{\mu\nu}^a)]$ h.c.	disagree	Same comments as the up-type scalar term: field content and charges are consistent through the H.c., but the EFT cutoff factor is not represented explicitly.
Paper assigns lepton numbers $L(\Phi_u) = L(\Phi_d) = -1$, $L(\Psi_u) = L(\Psi_d) = 0$ (Table X)	Reconstruction field table has no lepton-number column	missing-in-reconstruction	This is part of the paper's field definition, not just phenomenological commentary.

paper term (eq. ref)	reconstruction term	verdict	notes
Paper uses free masses m_{Ψ_q}, m_{Φ_q} (Eqs. 12–13)	Reconstruction parameter table lists defaults $M_{Fu} = M_{Fd} = M_{Su} =$ $M_{Sd} = 500$	extra-in-reconstruction	A benchmark/default mass may be implementation-specific, but it is not specified as part of the analytic model definition in Eqs. (12)–(13).
Paper coefficients are $\kappa_q^I/\Lambda_{\Psi_q}, \kappa_{qB}^I/\Lambda_{\Psi_q B}^3,$ $\lambda_q^{XI}/\Lambda_{\Phi_q}^2$ (Eqs. 12–13)	Reconstruction coefficients are complex C -parameters with real/imaginary parts and no displayed Λ	disagree	Complex couplings are compatible in spirit, but the reconstruction changes the coefficient bookkeeping and mass dimension unless the C 's are explicitly dimensionful Wilson coefficients.

Disagreements To Check

- **Missing scalar and fermion lepton numbers** — severity: **substantive**. A human should check whether the implementation tracks the accidental lepton-number assignments in Table X or intentionally omits them because they do not affect generated vertices.
- **All higher-dimensional operator coefficients lack explicit EFT cutoff powers** — severity: **substantive**. A human should check whether $C_{Fu}, C_{Fd}, C_{FBu}, C_{FBd}, C_{Su}, C_{Sd}$ are intended to be dimensionful Wilson coefficients already including $1/\Lambda, 1/\Lambda^2$, or $1/\Lambda^3$.
- **Scalar operators are written with $\Phi^\dagger$ and reordered barred spinors rather than the paper's $\Phi(q_R^c \sigma \ell_R)$** — severity: **convention**. A human should verify that the reconstruction corresponds exactly to the Hermitian-conjugate term included in Eq. (13), including coupling conjugation and color tensor conjugation.
- **Fermion operators are written as $\bar{\Psi} \Gamma q_R^c$ rather than the paper's $(q_R^c \Gamma \Psi)$** — severity: **convention**. A human should check the Dirac-charge-conjugation convention used by the implementation, especially signs for tensor bilinears.
- **Color tensor $\mathcal{C}$ is not named J and is expressed through K -type tensors** — severity: **convention**. A human should compare $\mathcal{C}$ against Eq. (A18), including the overall phase and index ordering, because a phase can be absorbed into couplings but an index swap cannot always be ignored.
- **Implementation default masses 500 appear in the reconstruction** — severity: **cosmetic**. A human should check whether these are merely UFO/FeynRules benchmark defaults rather than claims about the paper's physical model.

Overall Assessment

The reconstruction captures the main field content and operator inventory of the paper's Section III sextet model: two color-sextet Dirac fermions Ψ_u, Ψ_d , two color-sextet complex scalars Φ_u, Φ_d , the correct Standard Model gauge representations, and the intended $3 \otimes 6 \otimes 8$ quark-gluon-sextet interactions. Most apparent differences in chirality ordering, barred fields, conjugated scalars, and the color tensor notation look like implementation or Hermitian-conjugation conventions. The main physics-level gap is coefficient normalization: the paper's EFT structure is organized by explicit cutoff powers $1/\Lambda_\Psi, 1/\Lambda_{\Psi B}^3$, and $1/\Lambda_\Phi^2$, while the reconstruction replaces these with complex C -parameters without stating their mass dimension or relation to κ, λ, Λ .

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field's representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 22 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model's identity.

Physicist sign-off

- Reviewed by: _____ Date: _____
- Verdict: ☐ approve ☐ approve with corrections ☐ reject
- Notes: